1 Closure under full whiskering

For a finite graph H and $0 \le p < 1$, define

$$\Phi_H(p) := (1-p)^{v(H)} \chi_H\left(\frac{1}{1-p}\right).$$

Since χ_H is monic of degree $v(H)$, the right-hand side is a polynomial in $1-p$, and therefore extends continuously to $p = 1$. We use this continuous extension when $p = 1$.

Throughout, a graphon means a symmetric, jointly measurable function $W : \Omega^2 \to [0,1]$ on an arbitrary probability space $(\Omega, \mathcal{A}, \mu)$. Write

$$p = t(K_2, W) = \int_{\Omega^2} W(x,y)\, d\mu(x) d\mu(y)$$

and

$$d_W(x) := \int_{\Omega} W(x,y)\, d\mu(y)$$

for its degree function.

Definition 1 (Full whiskering)**.** *Let F be a finite graph with vertex set $V(F)$. The* full whiskering *of F, denoted by* $\mathrm{Wh}(F)$*, is obtained by adjoining, for each $v \in V(F)$, a new vertex v' and the single new edge vv'. Thus the newly added vertices are pairwise distinct and each of them has degree one.*

If $n = v(F)$, then

$$v(\mathrm{Wh}(F)) = 2n.$$

We first record the elementary path inequality needed in the proof.

Lemma 2. *Let W be a graphon of edge density p. Let P_4 denote the path on four vertices, hence with three edges. Then*

$$t(P_4, W) \ge p^3.$$

Equivalently,

$$\int_{\Omega^2} W(x,y) d_W(x) d_W(y)\, d\mu(x) d\mu(y) \ge p^3.$$

Proof. Write $d = d_W$. Since $W \ge 0$, if $d(x) = 0$, then

$$W(x,y) = 0$$

for almost every y. Tonelli and symmetry therefore show that the set of pairs (x,y) for which $W(x,y) > 0$ but $d(x)d(y) = 0$ has $(\mu \otimes \mu)$-measure zero. Hence, with the convention that quotients by $d(x) = 0$ are set equal to zero, we have almost everywhere

$$W(x,y) = \big(W(x,y)d(x)d(y)\big)^{1/3} \left(\frac{W(x,y)}{d(x)}\right)^{1/3} \left(\frac{W(x,y)}{d(y)}\right)^{1/3}.$$

Hölder's inequality therefore gives

$$\begin{aligned} p^3 &= \left(\int_{\Omega^2} W(x,y)\, d\mu(x) d\mu(y)\right)^3 \\ &\le \left(\int_{\Omega^2} W(x,y) d(x) d(y)\, d\mu(x) d\mu(y)\right) \left(\int_{\Omega^2} \frac{W(x,y)}{d(x)}\, d\mu(x) d\mu(y)\right) \left(\int_{\Omega^2} \frac{W(x,y)}{d(y)}\, d\mu(x) d\mu(y)\right). \end{aligned}$$

For the second factor,

$$\int_{\Omega^2} \frac{W(x,y)}{d(x)}\, d\mu(x)d\mu(y) = \int_{\{x:d(x)>0\}} \frac{1}{d(x)} \left(\int_\Omega W(x,y)\, d\mu(y) \right) d\mu(x)$$
$$= \mu\{x : d(x) > 0\} \le 1.$$

The third factor is bounded by 1 in the same way. Hence

$$p^3 \le \int_{\Omega^2} W(x,y)d(x)d(y)\, d\mu(x)d\mu(y).$$

Finally, integrating the two endpoints of a three-edge path first gives

$$t(P_4, W) = \int_{\Omega^2} W(x,y)d(x)d(y)\, d\mu(x)d\mu(y),$$

which proves the claim. □

The key observation is that full whiskering naturally changes the underlying vertex measure to the degree-biased measure.

Theorem 3 (Full-whiskering closure)**.** *Let F be a finite graph with*

$$n = v(F),$$

and let $p_0 > 0$. Assume the following two properties.

1. *For every graphon V on every probability space whose edge density s satisfies*

$$s \in [p_0, 1],$$

one has

$$t(F, V) \ge \Phi_F(s).$$

2. *The function*

$$s \longmapsto \Phi_F(s)$$

is nondecreasing on $[p_0, 1]$.

Then, for every graphon W of edge density

$$p = t(K_2, W) \ge p_0,$$

one has

$$t(\mathrm{Wh}(F), W) \ge \Phi_{\mathrm{Wh}(F)}(p).$$

More precisely,

$$t(\mathrm{Wh}(F), W) \ge p^n \Phi_F(p),$$

and

$$\Phi_{\mathrm{Wh}(F)}(p) = p^n \Phi_F(p).$$

Proof. Let

$$d(x) := d_W(x) = \int_\Omega W(x,y)\,d\mu(y).$$

Since

$$\int_\Omega d(x)\,d\mu(x) = p > 0,$$

the formula

$$d\nu(x) := \frac{d(x)}{p}\,d\mu(x)$$

defines a probability measure on $(\Omega, \mathcal{A})$.

Let $V(F) = \{1, \dots, n\}$. In the homomorphism-density integral for $\mathrm{Wh}(F)$, integrate first over the n newly added leaves. For a fixed value x_i of the original vertex i, its private leaf contributes the factor

$$\int_\Omega W(x_i, y)\,d\mu(y) = d(x_i).$$

Consequently,

$$\begin{aligned} t(\mathrm{Wh}(F), W) &= \int_{\Omega^n} \left(\prod_{ij \in E(F)} W(x_i, x_j) \right) \left(\prod_{i=1}^{n} d(x_i) \right) d\mu(x_1) \cdots d\mu(x_n) \\ &= p^n \int_{\Omega^n} \prod_{ij \in E(F)} W(x_i, x_j)\,d\nu(x_1) \cdots d\nu(x_n). \end{aligned}$$

Define

$$t_\nu(F, W) := \int_{\Omega^{V(F)}} \prod_{ij \in E(F)} W(x_i, x_j) \prod_{i \in V(F)} d\nu(x_i).$$

Thus

$$t(\mathrm{Wh}(F), W) = p^n t_\nu(F, W). \tag{1}$$

We next compute the edge density of W with respect to the probability measure ν. Put

$$p_\nu := \int_{\Omega^2} W(x,y)\,d\nu(x)d\nu(y).$$

Then

$$\begin{aligned} p_\nu &= \frac{1}{p^2} \int_{\Omega^2} W(x,y) d(x) d(y)\,d\mu(x)d\mu(y) \\ &= \frac{t(P_4, W)}{p^2}. \end{aligned}$$

By Lemma 2,

$$t(P_4, W) \geq p^3,$$

and hence

$$p_\nu \geq p. \tag{2}$$

In particular,

$$1 \geq p_\nu \geq p \geq p_0,$$

where $p_\nu \leq 1$ follows from $0 \leq W \leq 1$ and the fact that ν is a probability measure.

The same measurable kernel W is a graphon on the probability space $(\Omega, \mathcal{A}, \nu)$: changing the measure does not alter its measurability, symmetry, or pointwise range. Its edge density on this space is p_ν, and its F-density is exactly $t_\nu(F, W)$. The arbitrary-probability-space hypothesis for F therefore applies directly and gives

$$t_\nu(F, W) \geq \Phi_F(p_\nu).$$

Since Φ_F is nondecreasing on $[p_0, 1]$ and $p_\nu \geq p$, we obtain

$$t_\nu(F, W) \geq \Phi_F(p_\nu) \geq \Phi_F(p).$$

Together with (1), this yields

$$t(\mathrm{Wh}(F), W) \geq p^n \Phi_F(p). \tag{3}$$

It remains to identify the right-hand side with the chromatic-polynomial target for $\mathrm{Wh}(F)$.

Fix a positive integer k. Once a proper k-colouring of F has been chosen, each newly added leaf has exactly $k-1$ available colours, independently of all other leaves. Hence

$$\chi_{\mathrm{Wh}(F)}(k) = \chi_F(k)(k-1)^n.$$

Since this holds for every positive integer k, it is a polynomial identity:

$$\chi_{\mathrm{Wh}(F)}(x) = \chi_F(x)(x-1)^n. \tag{4}$$

Put

$$q := 1 - p.$$

For $p < 1$, using $v(\mathrm{Wh}(F)) = 2n$ and (4), we obtain

$$\begin{aligned}
\Phi_{\mathrm{Wh}(F)}(p) &= q^{2n} \chi_{\mathrm{Wh}(F)}(q^{-1}) \\
&= q^{2n} \chi_F(q^{-1})(q^{-1} - 1)^n \\
&= q^{2n} \chi_F(q^{-1}) \left(\frac{p}{q}\right)^n \\
&= p^n q^n \chi_F(q^{-1}) \\
&= p^n \Phi_F(p).
\end{aligned}$$

The same identity holds at $p = 1$ by continuity. Combining this with (3) gives

$$t(\mathrm{Wh}(F), W) \geq \Phi_{\mathrm{Wh}(F)}(p),$$

as required. □

Corollary 4 (Preservation of the chromatic threshold)**.** *Let F be a graph with*

$$r := \chi(F) \geq 3.$$

Assume that

$$t(F, W) \geq \Phi_F(p)$$

for every graphon W with

$$p = t(K_2, W) \geq 1 - \frac{1}{r-1},$$

and assume that Φ_F is nondecreasing on

$$\left[1 - \frac{1}{r-1}, 1\right].$$

Then

$$t(\mathrm{Wh}(F), W) \geq \Phi_{\mathrm{Wh}(F)}(p)$$

for every graphon W with

$$p \geq 1 - \frac{1}{r-1}.$$

Moreover,

$$\chi(\mathrm{Wh}(F)) = \chi(F) = r.$$

Thus full whiskering preserves both the Goodman-type inequality and the required threshold $p \geq 1 - 1/(\chi(F) - 1)$.

Proof. The density inequality follows from Theorem 3 with

$$p_0 = 1 - \frac{1}{r-1} > 0.$$

Since F is a subgraph of $\mathrm{Wh}(F)$,

$$\chi(\mathrm{Wh}(F)) \geq \chi(F) = r.$$

Conversely, take any proper r-colouring of F. Since $r \geq 2$, each new leaf can be assigned any colour different from the colour of its unique neighbour. Hence this colouring extends to a proper r-colouring of $\mathrm{Wh}(F)$, so

$$\chi(\mathrm{Wh}(F)) \leq r.$$

Therefore

$$\chi(\mathrm{Wh}(F)) = r.$$

□

Corollary 5 (Atlas 94)**.** *NetworkX Atlas graph* 94*, with graph6 code* `E@dW`*, is positive: for every graphon W of edge density $p \in [1/2, 1]$,*

$$t(F_{94}, W) \geq p^4(2p - 1) = \Phi_{F_{94}}(p).$$

Its order, size, and chromatic number are respectively 6*,* 6*, and* 3*, and*

$$\chi_{F_{94}}(x) = x(x-1)^4(x-2).$$

Proof. Let $F = K_3$. The accepted triangle (equivalently, length-three odd-cycle) bound states that, on every probability space,

$$t(K_3, W) \geq p(2p - 1) = \Phi_{K_3}(p) \qquad (1/2 \leq p \leq 1).$$

The polynomial $p(2p - 1)$ is nonnegative and strictly increasing on $[1/2, 1]$, since its derivative is $4p - 1 > 0$ there. Thus Theorem 3 applies with $n = 3$ and $p_0 = 1/2$.

In the labelling with triangle edges $01, 02, 12$ and private leaf edges $03, 14, 25$, the result is $\mathrm{Wh}(K_3)$. Atlas 94 has NetworkX edge list

$$04,\ 15,\ 23,\ 34,\ 35,\ 45.$$

The relabelling

$$0 \mapsto 4, \quad 1 \mapsto 3, \quad 2 \mapsto 5, \quad 3 \mapsto 0, \quad 4 \mapsto 2, \quad 5 \mapsto 1$$

carries the first edge list to the second. Hence Atlas 94 is exactly the full whiskering of the triangle. The chromatic identity gives

$$\chi_{F_{94}}(x) = x(x-1)(x-2)(x-1)^3 = x(x-1)^4(x-2),$$

and the density and target identities give

$$t(F_{94}, W) \geq p^3 \Phi_{K_3}(p) = p^4(2p-1) = \Phi_{F_{94}}(p).$$

At $p = 1/2$, the target vanishes (and the balanced bipartite graphon gives equality because the graph contains a triangle). At $p = 1$, $W = 1$ almost everywhere and both sides equal 1. □

This is exhaustive for the scoped Atlas catalogue on at most six vertices. A full whiskering in that range has a base of order at most three. If its result is non-bipartite, then the base is non-bipartite; among graphs of order at most three this forces a triangle component. Any additional isolated base vertex would become an excluded isolated edge component. Thus K_3 is the only eligible base and Atlas 94 is the only scoped full-whiskering row.

Remark 6 (Iteration)**.** *Suppose, in addition, that $\Phi_F \geq 0$ and is nondecreasing on the relevant interval. Since*

$$\Phi_{\mathrm{Wh}(F)}(p) = p^{v(F)} \Phi_F(p),$$

the same two properties hold for $\Phi_{\mathrm{Wh}(F)}$. Consequently the operation may be iterated.

If

$$F_0 := F, \qquad F_{j+1} := \mathrm{Wh}(F_j),$$

then

$$v(F_j) = 2^j v(F)$$

and

$$\Phi_{F_j}(p) = p^{(2^j-1)v(F)} \Phi_F(p).$$

Thus every iterated full whiskering satisfies the same Goodman-type inequality on the same edge-density interval.

1.1 Inputs, hidden-assumption audit, and exact limitations

The general closure theorem uses only Tonelli–Fubini, the three-edge path inequality proved above, degree-biasing of the original probability measure, and the assumed inequality and monotonicity for F. Corollary 5 additionally uses exactly the accepted triangle bound. No regularity, minimum-degree hypothesis, finite-step representation, or injective-copy interpretation is used. Points with $d_W(x) = 0$ are handled by an almost-everywhere convention justified before Hölder is applied, and the proof never divides by $d_W(x)$ pointwise outside that convention.

The hypothesis $p_0 > 0$ is essential to the degree-biased probability measure as written. It is automatic for the required threshold when $\chi(F) \geq 3$. The endpoint $p = 1$ is handled by the

polynomial continuation and directly by the same measure-change proof; no substitution of $1/(1-p)$ is made there.

Full whiskering means one distinct private leaf at every vertex of F. The theorem does not cover a leaf at only some vertices, several leaves at a vertex, shared leaves, or arbitrary pendant-tree attachment. It also requires monotonicity of Φ_F on the whole density interval; positivity of F alone does not imply that monotonicity. Consequently this note is not a general leaf-attachment closure theorem.

A What the Lean formalization actually proves

Corollary 5 is formalized: Atlas 94 is `verified`, as `satisfiesLowerBound_94` in `lean/Taeyoung/Methods/Whisker`
One difference of substance.

Theorem 3 is not stated in general

The change of measure is formalized in full and in general (`Methods/DegreeBias.lean`): $d\nu = (d_W/p)\,d\mu$ is a probability measure, W reread on (Ω, ν) is a graphon, its densities are the degree-weighted integrals (1), and (2) — the biased edge density is at least p, by Lemma 2 — is `le_cliqueDensity_degreeG`
That is the whole analytic content of the theorem.

What is not formalized is the combinatorial half: $\mathrm{Wh}(F)$ is not constructed as a graph operation, so the identity $t(\mathrm{Wh}(F), W) = p^n\, t(F, W_\nu)$ is not available for a general F. It is supplied per row, by the peeling identity for that row's own graph, run on both sides — six coordinates and three for Atlas 94. The hypotheses of Theorem 3 are then discharged concretely: Goodman for $F = K_3$, and monotonicity of $z \mapsto z(2z-1)$ on $[1/2, 1]$ in place of the abstract hypothesis (2).